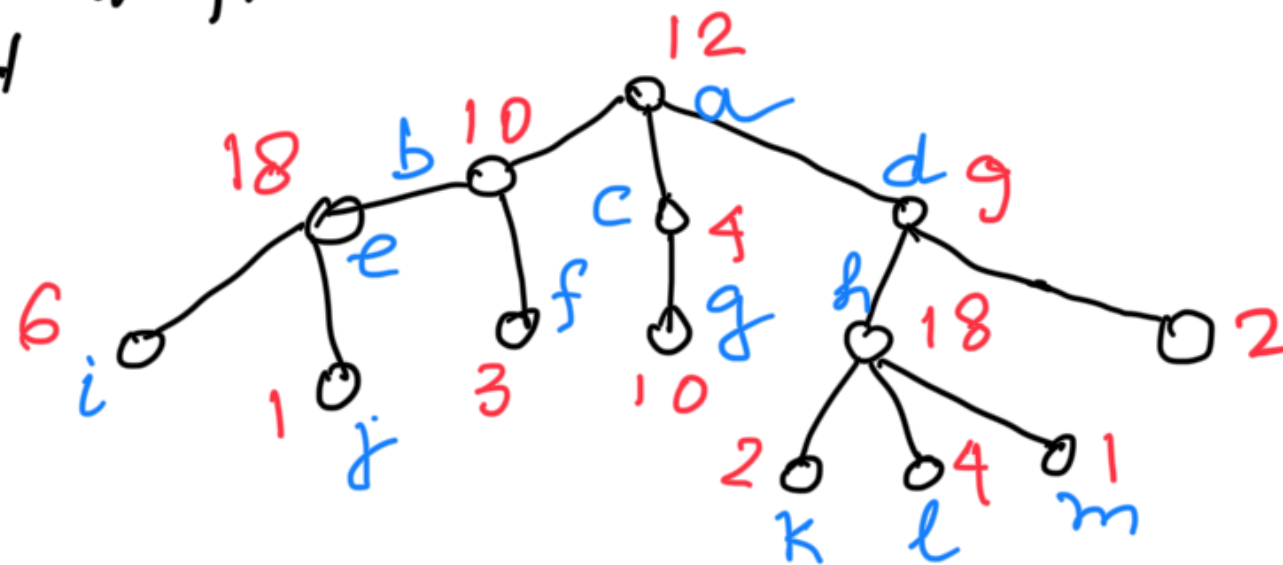


Goal is to find
the highest
wt.
Independent
subset



Notation a: node
 a^* : subtree rooted
at a

Choose $\rightarrow$

$$\max \left\{ 12, 10 + 4 + 9 \right\} \text{ or } \max \left\{ 12, \max b^* + \max c^* + \max d^* \right\}$$

$+ \text{best soln of the grandchildren of } a$

$\max i = \max i^* = 6$

$\max b^* = \max \left\{ 10, \text{best soln from children of } b \right\}$

a is chosen $\rightarrow$ $+ \text{g children of } a$

a is not chosen $\rightarrow$ $\text{best of the children of } a$

Let $P(i)$ denote the highest weight independent subset

$P(u)$ is the highest wt independent subset in u^* (u^* is the subtree rooted at u)

$$P(u) = \max \left\{ wt(u) + \sum_{v \in \text{graph}(u)} P(v), \sum_{w \in \text{child}(u)} P(w) \right\}$$

$$= wt(u) \text{ if } u \text{ is a leaf node}$$

Computation time : Using a charging argument on the nodes (once as g.c. and once as a child),

the total # changes = $O(\# \text{ nodes}) = O(n)$

Suppose tree is unweighted — is there a simpler algo?

Shortest path in graphs

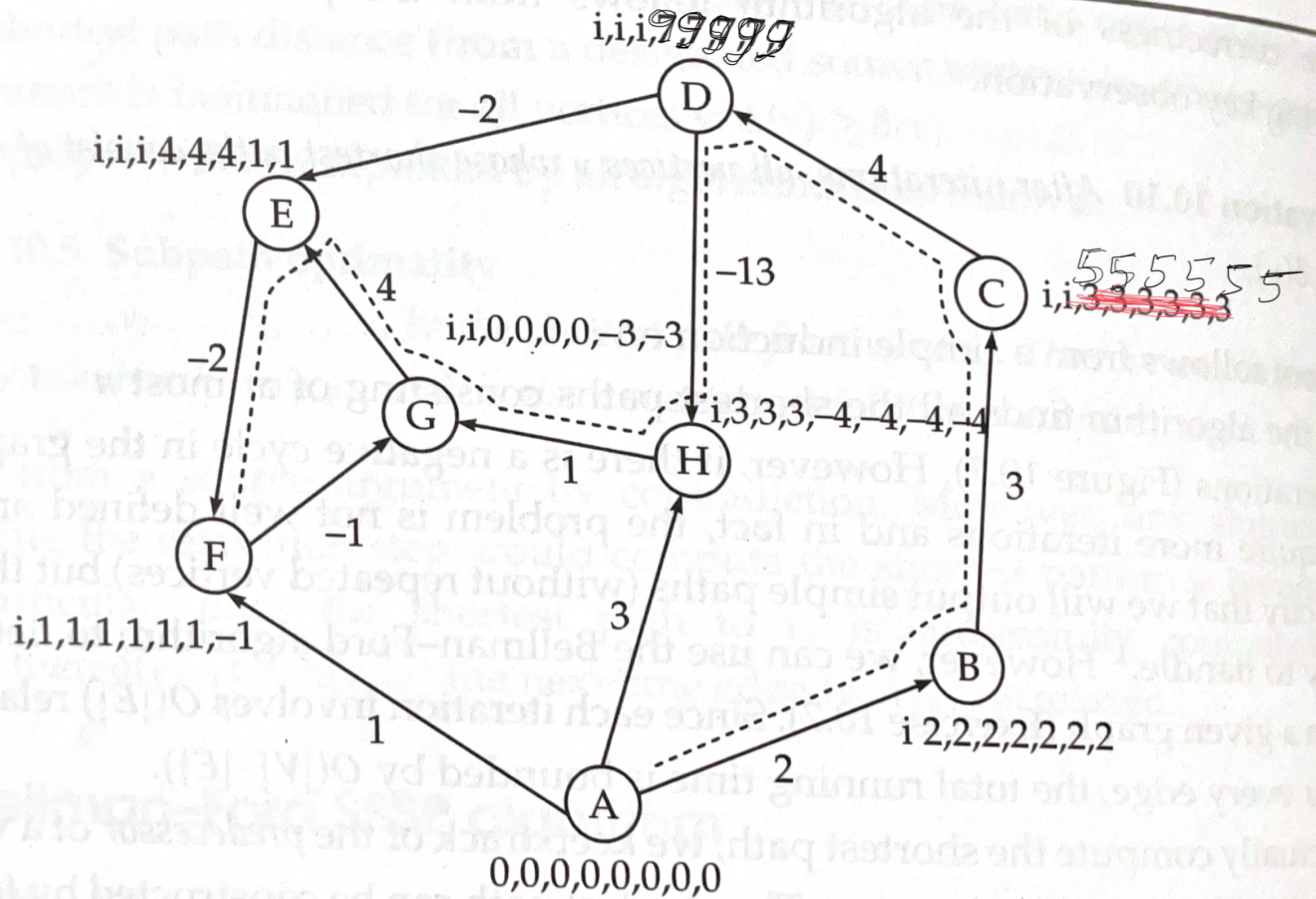


Figure 10.7 For every vertex, the successive labels over the iterations of the Bellman-Ford algorithm are indicated where *i* denotes

∞ . The dotted line shows the final path computed for vertex F .

Shortest path versions

1. single starting $\rightarrow$ single destination
2. single source $\rightarrow$ all destinations SSSP
3. all sources $\rightarrow$ all destinations APSP

any
wts

non-negative wts

Properties of shortest paths

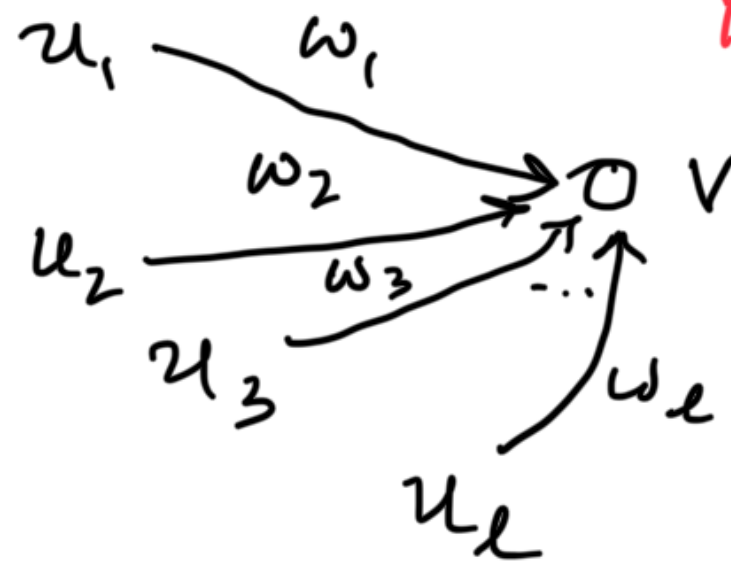
1. Subpath optimality

$$s = v_0 \rightarrow v_1 \rightarrow v_2 \sim \dots \sim v_i \sim \dots \sim v_k = t$$

any subpath is shortest

Notation $\delta(v)$: shortest path to v

2.



$$\delta(v) = \min_i \{ \delta(u_i) + w_i \}$$

$\delta(v) \leq \Delta(v)$ $\Delta(v)$: upper bound on shortest path length from s to v

$\text{Relax}(u_i, v)$: If $\Delta(v) > \Delta(u_i) + w_i$ then

$$\Delta(v) = \Delta(u_i) + w_i$$

Idea behind any shortest path algorithm is to start with a lower $\Delta(v)$ - as

and keep refining till $\Delta(v) = \delta(v)$

Initialise $\Delta(s) = 0$ and $\Delta(v) = \infty$
for $v \neq s$

while some $\Delta(v)$ changes repeat
[for all edges $(x, y) \in E$
Relax (x, y) time $O(1)$ for each edge
end $\Rightarrow O(|E|)$ for each iteration

Correctness ?

Complexity ?

Claim: If there exists a shortest path from $s \rightsquigarrow v$ that consists of $i < n$ edges, then $\delta(v) = \Delta(v)$ after i iterations

$\Rightarrow O(|V| \cdot |E|)$ algorithm

Proof by induction on the # edges in shortest paths

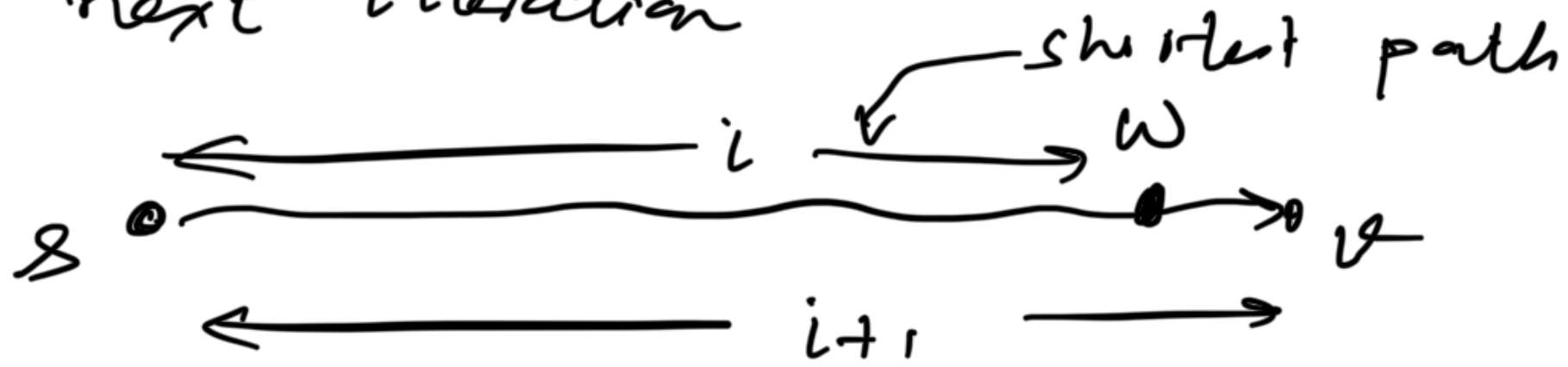
$i = 0$

$\delta(s) = \Delta(s)$ as per initialization

No -ve cycles

Suppose true for all ^{shortest} paths that have i edges

Consider what happens to the shortest paths having $i+1$ edges in the next iteration



$\Delta(w) = \delta(w)$ after i iterations as per I. 7.1.

$\Rightarrow$ Next iteration $\Delta(v) = \delta(v)$
after relaxing (w, v)

What about negative cycles

Let $\delta_i(v)$ denote shortest path to v

δ consisting of $\leq j$ edges
 Then $\delta(v) = \min_{v \neq s} \min_{1 \leq j \leq n} \delta_j(v)$ ($= \infty$ if there is no path of $\leq j$ edges)

Prove that $\Delta(v) = \delta_j(v)$ after j iterations